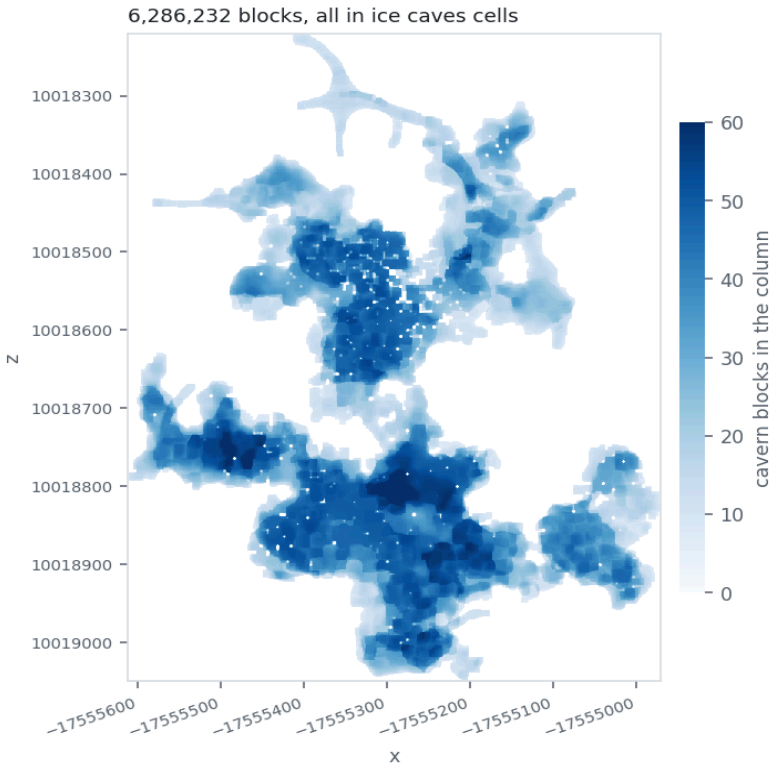
The largest ice caves cavern: 6.29 million blocks

Ice caves arrived in Minecraft 26.4-snapshot-3 on 6 October 2026. The task, the same day: find the largest cavern in them, preferably one that is ice caves throughout. The best so far is in seed **1844768299621927670** at `/tp -17555309 -37 10018737`: **6,286,232 blocks** of connected cavern, every one of them in an ice caves cell, confirmed on the terrain the snapshot's own code generates and by its own biome source. The search ran on one 12-thread laptop and, for its last hours, also on a rented 32-thread machine whose graphics card does the seed gate ($1.45 of rent in all).



The record cavern from above: cavern blocks per column.

1. What is measured

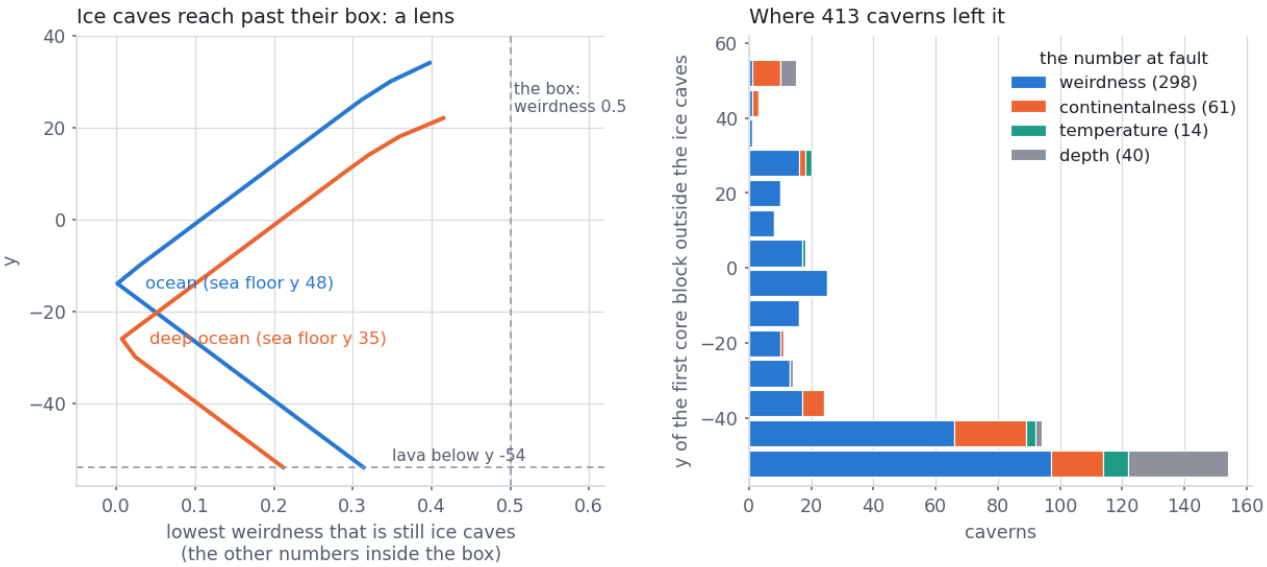
A cavern as in the cave search: underground air within 3 blocks of a connected core of 7×7×7 air cubes, noise terrain and aquifers only. A block is in ice caves when its 4×4×4 biome cell is. Two categories:

- **ice caves throughout**: every block of the cavern;
- **ice caves volume**: the cavern's blocks in ice caves cells, whatever the rest is.

The snapshot's rule, read from its jar: ice caves are a cave biome (depth 0.2 to 0.9) at temperature -1 to -0.8, humidity up to 0.7, continentalness -1.05 to -0.19 (the oceans) and weirdness 0.5 to 1.1. The terrain code is 26.3's, so the cave search's exact generator still holds.

2. What an ice cavern is

The game gives a cell the biome whose box is *nearest*. Underground the surface biomes are far away in depth, so ice caves reach past their box by up to min(depth, 1 - depth): a lens under a cold ocean. At y -16 a weirdness of 0 still counts; at the roof and at the floor almost nothing outside the box does. The lowest air, y -53 over the lava, is at depth 0.81 under an ocean, so only weirdness over 0.31 is ice caves there.



Left: the lowest weirdness that is still ice caves, by height, under the two kinds of ocean. Right: where the core of 413 caverns first left the ice caves in an exact test, and which climate number was out.

Three caverns in four that leave the ice caves leave through the floor. Air is dry only well under the sea floor, so an ice cavern is a slab between the lava and about y 20, some 22 blocks thick, and its size is its footprint. **A cavern that is ice caves throughout is a cheese cave closed off inside one weirdness lobe.** That sets what every stage has to look for: a lobe that is large, cold and under ocean, with open cave inside it and rock at its rim.

3. The search, stage by stage

stage	what it decides	cost
phase gate	are ten noise layers where they should be in y: temperature and weirdness half way between their lattice planes at y 0, the two lowest cheese layers strongest near the cavern floor? (RNG steps only, stops at the first layer outside its window)	30 ns a seed on a laptop thread; 0.06 ns on a rented RTX 4060
scanner: pairs	32 million cave-noise alignments on 6-step windows, of which only those with a place whose mean weirdness over the window is 0.40 or more go on (three table lookups: no sample of the world)	0.4 s
scanner: climate	which of the 1.7 million places with that weirdness are under ice caves climate?	1.0 s
scanner: first look	coarse clusters under an ice caves mask, at every such place (about 22,000)	1.6 s
scanner: fine look	clusters on 32-block tables with the ice caves column by column, at the best 1,300 by a fitted index + 200 by ice points	1.9 s
formula	the game's densities on a 32-block lattice, at the 48 candidates most likely to be pure: is the cluster ice caves throughout, and how much of its rim is not?	1.0 s
exact	the cavern block by block, for pure clusters of 450+ samples; given up the moment its core leaves the ice caves	2.9 s

(Seconds a seed on one thread of the rented box with all 32 busy: 9 thread-seconds a seed with the tables' 0.9 s. The laptop takes 12.)

4. The ideas, in the order they came

A sweep of one world (dropped). The first attempt walked seed 1 square by square. It found 1.95M in 106 squares and was the wrong tool: the cave search had shown that a world's best places can be listed without visiting it.

The cave scanner with a climate test (build D). The cave noise of a world is two small tables, and each pair of low windows sits at 2 or 3 places per axis. The scanner tested the climate at every place of every pair and scored the rest as the cave search did. 23 thread-seconds a seed; 2 caverns of 2M+ in the 96 test seeds.

Measure each stage, and find that the cave search's order does not carry over. With every place of 64 seeds evaluated in full, a pair's mean cheese and its open points, which rank giants well, say nothing about which places hold a cavern that is ice caves throughout. What does: a coarse cluster count *under an ice caves mask* (temperature and continentalness at 16 points, weirdness on 8×8, by depth). The best 1,000 of 26,000 places by it hold 63 to 77% of the good ones; the old order held 12%.

A cheaper climate test. The temperature's wide octave is nearly a plane across a window, so its value and slope in the middle (two noise samples) bound it at all 16 quarter centres; five windows in six go there. Cold quarters get the weirdness first. 7.7 million places in 1.2 s instead of 2.7.

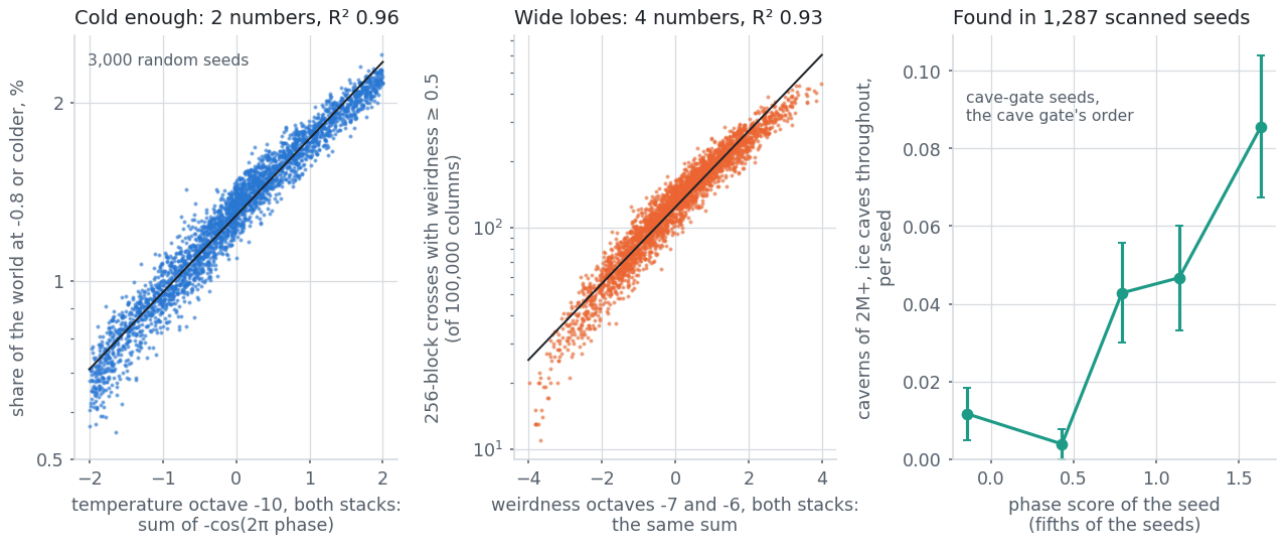
Why big clusters failed the exact test. Clusters the formula called pure passed the exact test 5 to 10% of the time once they were large. Making the test report *where* a cavern leaves gave the figure above: through the floor, by the weirdness. The fine look and the formula now sample at y -53, apply the terrain's bottom slide, count a point as ice caves only if the cells at both ends of the blocks it stands for are, and count a cluster's **edge**: sideways neighbours that are rock or water outside the ice caves. Clusters with no edge pass 81%.

Let the nearly pure through (build E2). The stricter rule marked a few points of true caverns as outside. A recall count over 112 known caverns showed where each was lost; letting clusters with at most 3% outside through to the formula found 36 of the 46 known caverns of 1.4M+ in the test seeds, where the build before found 7.

build, same 96 seeds	thread-s a seed	ice caves throughout, 1.4M+ found	2M+
D: cave scanner with a climate test	23	10	2
E: fast climate test, first look, new fine look	13	7	4
E2: nearly pure clusters let through	17	36	8

Measure only what can be a record. With every candidate of 192 seeds tested, the exact test on pure clusters of 270+ samples alone keeps 12 of the 13 caverns of 2M+ for a quarter of a test a seed instead of three. 12 thread-seconds a seed.

The gate (after being asked whether the cave search had been read at all). It had not been, and its first lesson applied directly. A Perlin layer is weakest on its lattice planes and strongest half way between, and the climate is only ever sampled at y 0, so one random number per layer decides how far that layer swings anywhere in the world. On 4,000 random seeds two numbers explain 96% of how much of a world is cold enough, and four more 93% of how many wide weirdness lobes it has. Between worst and best phases the count of 128-block lenses under ice caves climate changes a hundredfold; beyond the phases, seeds differ by 6 to 10%.



Left and middle: what the phases of six layers predict, on 3,000 random seeds. Right: caverns of 2M+ actually found in cave-gate seeds scanned in the cave gate's own order, by fifths of the phase score.

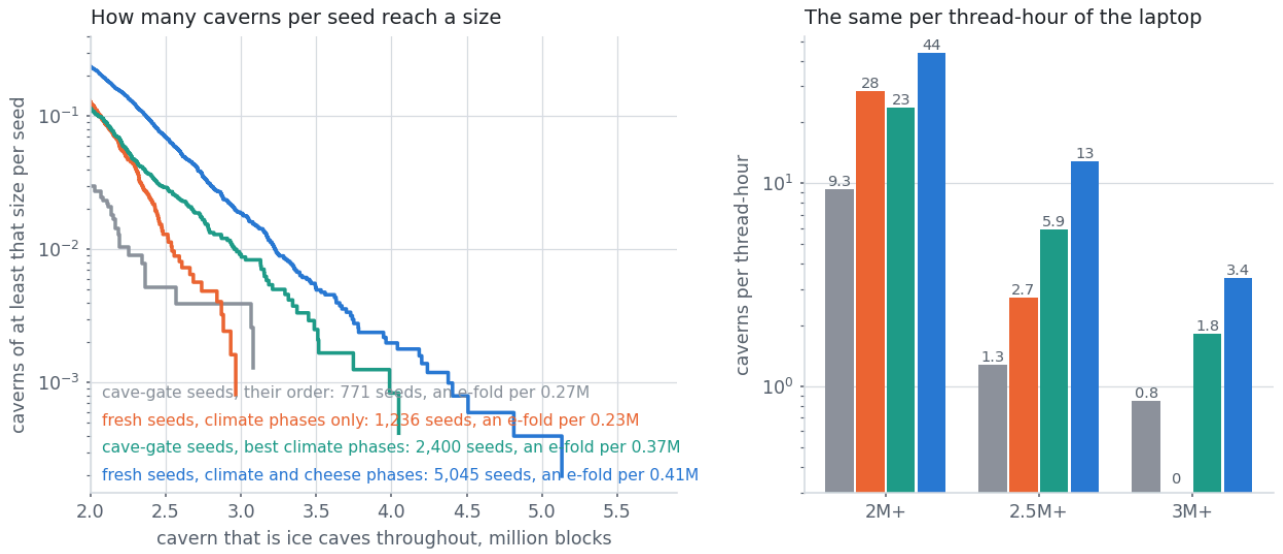
layer, both stacks	**effect on the log rate per unit of -cos(2π phase)**
temperature octave -10	0.29 on the cold share; 0.32 ± 0.14 on caverns of 2M+
weirdness octave -7 (512-block cells)	0.37 on lobes 256 wide, about 1.0 on lobes 512 wide; 0.63 ± 0.20 on caverns of 2M+
weirdness octave -6	0.37 on lobes 256 wide
temperature octave -8	-0.25 ± 0.13 on caverns of 2M+; tied to octave -10 by the seed generator (below), so not a free choice
continentalness, humidity, cave_layer, aquifer floodedness	nothing measurable

The cave gate's 330,000 seeds were then scanned in the order of that score instead of their own, which tripled the rate of caverns of 2M+ and raised that of 3M+ about eightfold.

Layered gates on fresh seeds (after "you can layer gates"). The cave gate's seeds are a fixed list. Fresh seeds need their own cave-side gate, and the first reckoning said that was too dear for a laptop: tight windows on climate *and* cheese pass one seed in billions. The measurement said otherwise. Each phase test stops at the first layer outside its window, so a gate costs its first test times the product of its windows, and what matters is how that product is shared out. With the climate windows loosened and two more layers added in front of the scanner (the two lowest cheese octaves at the caverns' height), fresh seeds beat the best of the cave gate's:

seeds	scanned	thread-s a seed (gate + pipeline)	2M+ per seed	2.5M+	3M+	2M+ per thread-hour	2.5M+	3M+
cave-gate seeds, their own order	771	0.0 + 11	0.029	0.004	0.003	9.3	1.3	0.8
cave-gate seeds, best climate phases	2,400	0.0 + 17	0.114	0.029	0.009	23.5	5.9	1.8
fresh, climate phases only	1,236	0.2 + 16	0.126	0.012	0.000	28.1	2.7	0.0
fresh, climate tight + cheese loose	304	7.2 + 14	0.260	0.053	0.013	43.3	8.8	2.2
fresh, ten windows by weight	720	0.7 + 18	0.257	0.042	0.007	50.3	8.2	1.4
fresh, twelve windows (with pillars)	890	1.7 + 16	0.315	0.067	0.020	62.6	13.4	4.0
fresh, ten windows, refit centres (production gate)	1,277	1.7 + 25	0.269	0.081	0.020	36.7	11.1	2.7
fresh, fourteen windows (entrance and pillar layers added)	342	1.7 + 21	0.105	0.044	0.006	17.1	7.1	0.9
production gate, half the pairs by own weirdness	3,229	1.7 + 15	0.149	0.069	0.018	32.5	15.0	3.9

(Caverns that are ice caves throughout; one measuring policy for all rows.)



Left: caverns that are ice caves throughout per seed, by size, for four sources of seeds. Right: the same per thread-hour of scanning them.

Climate phases alone give as many caverns of 2M+ as the cave gate's seeds and none of 3M+ in 1,236 seeds: the cave noise sets the ceiling. Two rules came out of fitting what the fresh seeds yielded (`gatefit.py`):

- *Windows by weight.* If a layer's phase is worth w in log rate, the box of windows that loses least for a given share of seeds kept has half-widths proportional to 1/√w. The first cheese test had the climate far too tight and the cheese far too loose; sized by weight, the same yield per seed cost a tenth of the scanning.

- *Weights grow with the target.* The lowest cheese octave is worth 0.4 per stack for clusters of 270+ samples, 0.7 for 350+ and 1.0 for 430+, and its best height is lower than the slab's middle: the noise should peak near the cavern floor (mid-cell at about y -30 for octave -8). The pillar noise, the cave-layer noise and the aquifer floodedness show effects of 0.2 to 0.3 on the largest clusters only; they are not gated until more data confirms them.

Phases that the seed generator ties together. A gate with fourteen windows passed a tenth of the seeds it should have. The reason is in how the game seeds a noise: every octave's generator is the stack's state XOR a constant, and the generator's step is linear in XOR, so some octaves of one stack get y offsets that move together. On 200,000 seeds: weirdness octave -5 sits half a cell from octave -6 (within 0.1), so -6 at mid-cell forces -5 onto its plane; temperature octave -8 mirrors octave -10 (mid-cell with mid-cell); cheese octave -6 follows octave -8 within 0.19. Octaves -8 and -7, and -7 and -6, are independent, as are the two stacks. A tied layer cannot be chosen, so it stays out of the gate, and a weight fitted for it belongs partly to its partner (the "negative" effect of temperature octave -8 above is that).

A gate that was worse. The fit over 25 layers also flagged the cave-entrance noise (z 3.9) and the pillar noise. A fourteen-window gate with both found half as many caverns of 2.5M+ per seed as the ten-window one (0.040 against 0.081): a hint from a scan of many layers is not evidence, and every unconfirmed window widens the real ones. A new layer now gets its own arm before it goes into the gate.

How many spots per seed ("isn't this still scanning whole seeds?"). The scanner takes a seed's million best cave-noise pairs. Fewer is not better: the best 100,000 hold 16% of the large places of the best million, 250,000 hold 32%, two million 125%, while a seed's tables, gate and fine look cost the same at any depth. A giant of the cave search sat at its seed's very best alignment; here the cave noise only has to be good, and what decides is whether a good alignment falls under a large cold lobe, which 0.6% of the places do. (A search at spawn only, one place per seed, would for the same reason be three to four orders of magnitude slower for the same size.)

Rank places or caverns? ("a composite index"). A logistic index fitted on every place under ice of 39 seeds (0.40 × ice points - 0.27 × all points of the coarse cluster) holds 71% of the large places in its best 1,050, where the two lists in use held 45%. In a paired run on 300 seeds it found 25 caverns of 2.5M+ where the lists found 24: a cavern shows in several overlapping windows, and one of them was already on a list. From there every pruning idea was judged by **recall of known caverns** (`recall.sh`: the scanner alone on the seeds of 133 known caverns of 2.7M+, counting those it still lists):

scanner setting	2.7M+ kept	3M+ kept	3.5M+ kept	scanner, s a seed (busy machine)
two lists, 700 + 350 fine looks (as before)	125 of 133	53 of 55	13 of 13	17
index, 900 + 150 fine looks	127 of 133	53 of 55	13 of 13	19
index, 300 + 50	105 of 133	47 of 55	12 of 13	16
index, 100 + 20	72 of 133	35 of 55	9 of 13	14
index, 300 + 50, half the pairs dropped by own weirdness	95 of 133	47 of 55	13 of 13	10
index, 300 + 50, two thirds dropped	92 of 133	44 of 55	12 of 13	8
index, 600 + 100, two thirds dropped	102 of 133	46 of 55	12 of 13	10
index, 600 + 100, four fifths dropped	86 of 133	40 of 55	11 of 13	8
index, 600 + 100, two million pairs, three quarters dropped	93 of 133	44 of 55	12 of 13	13
index, 300 + 50, first look only where 2+ quarters are in the box	109 of 133	46 of 55	10 of 13	12
index, 300 + 50, first look only where 3+ quarters are in the box	78 of 133	38 of 55	7 of 13	10
index, 300 + 50, the best half million pairs by cave noise	59 of 133	27 of 55	6 of 13	9

Part of the biome belongs to the pair (“surely there’s a cheaper way to find the intersection”). The weirdness’ octaves -6 and -5 repeat with exactly the cave noise’s period, so their mean over a pair’s window is the same at every one of the pair’s places: a table lookup per pair, before any place is visited. Dropping the pairs with the lowest “own weirdness” loses almost none of the largest caverns and cuts the climate test and the first look in proportion.

The whole weirdness is a table (“why not prune more?”). That filter used two of the weirdness’ three octaves. The third (cells of 512 blocks) repeats every *two* periods of the cave noise, and a pair’s places are 331 periods apart, an odd number. So a pair’s places fall into 2 x 2 classes, and for each class the mean weirdness over the middle of the window is exact from tables. On the 133 known caverns, the 15% of places with the highest such “box weirdness” kept what the pair filter kept at 50%. Tested inside the loop that picks the pairs, from memory read in order, a dropped pair costs about a nanosecond, so the cave noise can be taken 16 to 32

times deeper for the same time. Evaluating every place under ice of 36 seeds at 16 million pairs showed why that pays: the cave-noise rank says nothing about the yield of a place down to the 16-millionth pair, while the yield rises sixfold from a box weirdness of 0.25 to 0.50. A seed holds about 96 near-pure clusters of 400+ points; the scanner had been listing 9.

Smaller windows. With depth free, the window the pairs are picked on should be the size of an ice cavern, not of the largest cave: 12 and 16 steps (780 and 1,040 blocks) dated from a million pairs a seed. And then the formula stage was the limit (next paragraph). The whole pipeline on the same seeds:

setting	thread-s a seed	2M+	2.5M+	3M+	3.5M+
laptop, 200 seeds: 1M pairs, half by their own weirdness, windows of 12 and 16 (before)	8.1	22	6	3	2
16M pairs, box weirdness 0.35+	9.6	98	54	16	2
the same, 96 formula candidates by size	12.0	126	63	17	3
windows of 12 only, 48 candidates by the index	11.5	146	76	25	5
windows of 10	12.9	184	102	37	12
windows of 8	14.6	281	172	67	16
box, 651 seeds: windows of 10	10.0	653	348	106	27
windows of 8	12.2	885	551	203	46
windows of 6	13.4	1,166	807	292	66
windows of 4	14.0	1,223	921	307	55
windows of 6, 32M pairs, box weirdness 0.40+ (in use)	13.6	1,184	835	316	64

(Caverns that are ice caves throughout; the exact test from 350 samples in every row.)

Rank what goes to the formula. 48 candidates a seed went to the formula by cluster size, and 2% of them had a pure cluster of 350+ samples in the game's densities. The share of the fine cluster outside the ice caves predicts that far better than its size does (0 to 1% outside: 6.6%; 6 to 8%: 0.6%). On held-out seeds the best

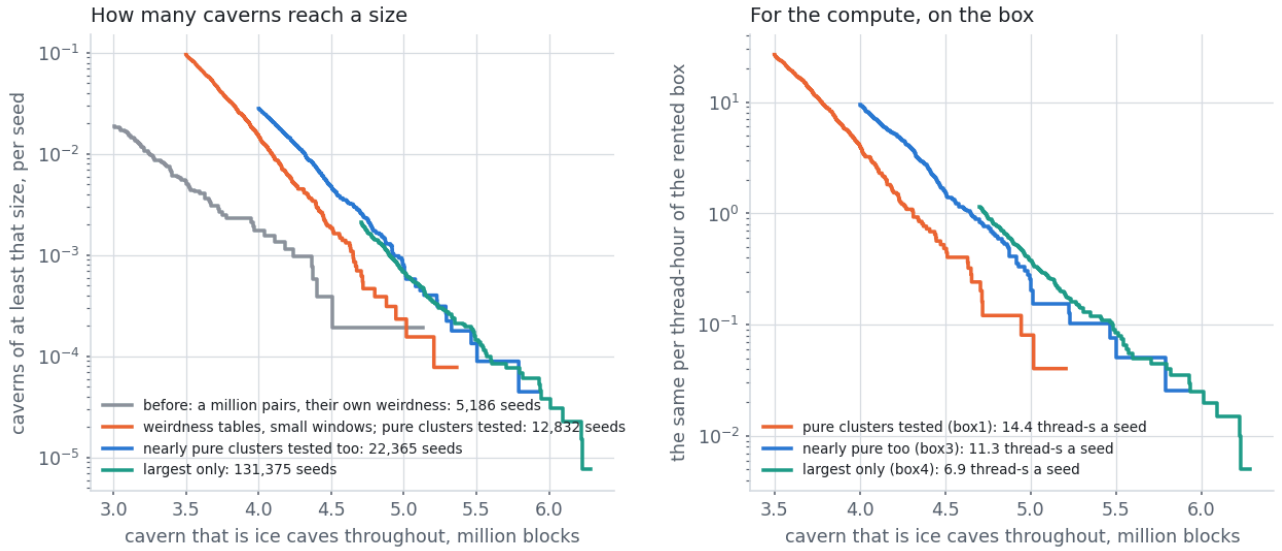
tenth of the candidates by 3.1 ln(size) - 0.54 x (percent outside) held 330 of 674 such clusters; by size, 69. 48 candidates by it find what 96 by size did. (That index was for small pure clusters, and it had to go again: see the next paragraph.)

A leak at the top (found by asking whether the new settings still find the old record). They did not. Of the 32 caverns of 3.6M+ known from before, the new pipeline found 25, and the 5.13M one was not among them: its cluster in the game's densities has 792 samples of which the formula's cautious rule puts 11 outside the ice caves, so it is not "pure", and it had only ever been measured because its 781 ice samples passed the ice-volume list's threshold of 780. The strict test now also goes to the largest nearly pure clusters (600+ samples, at most 2% outside, and either at most 4 outside or 700+ in all). On 651 seeds 476 such clusters gave 24 caverns that are ice caves throughout, 10 of them 3.5M+ that the test of pure clusters does not reach (13% pass with 1 or 2 samples outside, 2% with 8 to 11). In production its first, narrower version took 0.36 tests a seed against 1.98 for pure clusters and found 3 of the 4 caverns of 5M+ and 11 of the 21 of 4.5M+ of 4,660 seeds: the top of the tail lives in nearly pure clusters, and the caverns of 4M+ a seed rose from 0.016 to 0.028. With the target changed, the ranking for the formula was wrong too: with the formula run on every listed candidate of 651 seeds, 48 runs a seed by size hold 95% of the clusters the exact test now wants and all of the largest, where the fitted index held 72% and 82%. Which window size is used matters little at the top: seven settings together found 14 caverns of 4M+ in those seeds, the one in use 12.

A card for the gate, a box for the pipeline ("$1 for a proper GPU run"). The gate is integer arithmetic on the seed's random numbers. On a rented RTX 4060 it looks at 1.7 x 10^{10} seeds a second, 560 laptop threads' worth, and prints exactly the CPU gate's seeds (124 of 124 on a range of 6 billion; every hit is checked again by the CPU code). That affords windows a third narrower (4 x 10^{-10} of the seeds kept instead of 2 x 10^{-8}) and leaves every core to the pipeline: 31 workers on the box's i9-14900KF do 13,000 seeds an hour, four times the laptop, and find the same caverns as the laptop on the same seeds. A guard on the laptop copies the results every ten minutes and, at the spending cap, stops the run, downloads and checks everything, and only then destroys the instance. In ten and a half hours and $1.45 the box scanned 130,108 seeds and found 4 of the 5 caverns of 6M+.

policy	seeds	4M+ per 1,000 seeds	4.5M+	5M+	5.5M+	e-fold above 4M
before: a million pairs, their own weirdness	5,186	1.7	0.39	0.19	0.00	0.37M +- 0.12
weirdness tables, small windows; pure clusters tested	12,832	15.2	1.87	0.23	0.00	0.23M +- 0.02
nearly pure clusters tested too	22,365	28.2	4.83	0.80	0.13	0.30M +- 0.01
largest only	131,512			0.68	0.14	

(Caverns that are ice caves throughout; the counts of the last row are complete from 4.7M only. On the box a seed takes 11 thread-seconds with the fuller policies and 7 with the largest-only one: 10,000 and 16,000 seeds an hour for $0.14.)



Left: caverns that are ice caves throughout per seed, by size, for the four policies in turn; each curve starts where its counts are complete. Right: the same per thread-hour of the box, for the three policies it ran.

The runs ended on 7 October at 12:45 UTC with the box's budget. To go on: the gate on two laptop threads feeding ten workers (`gaterun.sh NAME spec_p1.txt 2 10`), or on a card (`fleet/`), with `seed.sh`'s defaults (complete from 4M) or the largest-only settings of section 6.

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5. Results

179,260 seeds scanned in 504 thread-hours (a 12-thread laptop and, for the runs named box, a rented 32-thread machine). Volumes are from the native measure (`icemeasure`). "Checked" is what was confirmed with the snapshot itself: *terrain* = the same volume on the terrain the snapshot's code generates (`measure_snapshot.sh`), *biome* = the snapshot's biome source asked for every cell of the cavern (`verify_ice.py`).

Ice caves throughout

Every block of the cavern is in an ice caves cell.

blocks	seed	/tp	checked
6,286,232	1844768299621927670	-17555309 -37 10018737	terrain and biome
6,229,370	8382283966336609321	20778699 -31 22572451	terrain and biome
6,225,152	-7523541043407331934	-18620141 -10 -29807944	terrain and biome
6,091,533	-7981004725191640799	8037267 -28 8863065	terrain and biome
6,013,285	-2038430933008404415	10005539 -41 -25054085	terrain and biome
5,945,554	7721026526635531331	13443965 -23 -27238214	biome
5,936,449	-8129084442917290139	-17008755 -22 -1746372	biome
5,930,365	608929704024686983	-4591226 -28 -2615810	biome
5,928,697	7617029319468386338	18596670 -24 12316442	terrain and biome
5,820,486	2601411861908076280	-15301409 -35 5709594	biome
5,797,211	-7280165585465947950	-4508080 -27 -18476567	biome
5,791,476	7248708726046152183	1115181 -28 -5111265	terrain and biome

38292 of 2M+, 32337 of 2.5M+, 22429 of 3M+, 10716 of 3.5M+, 3521 of 4M+ in all.

All but a few blocks

The core is ice caves throughout; the blocks outside are in the 3-block rim.

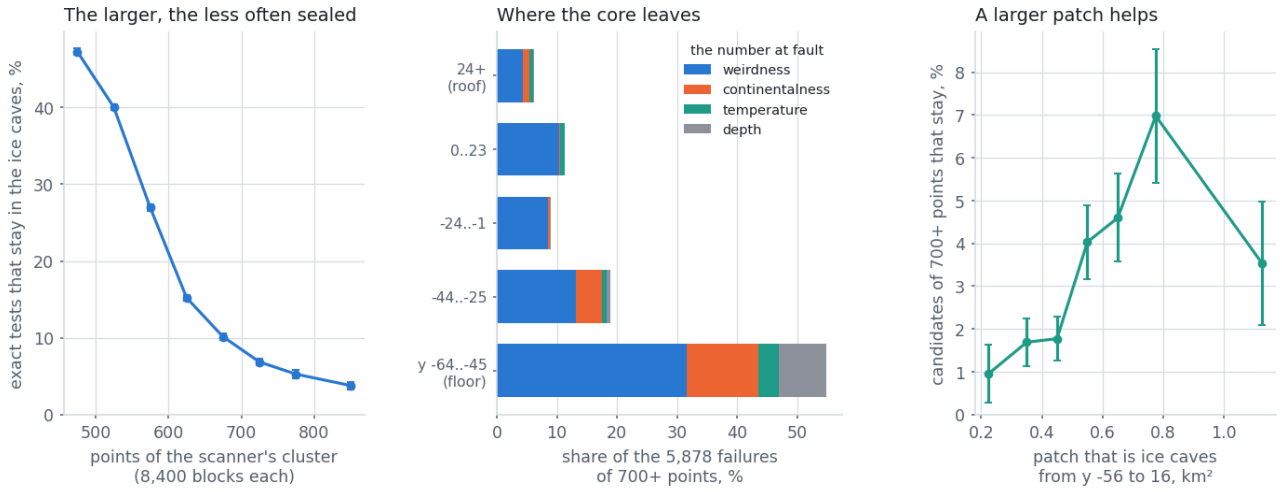
blocks	outside ice caves	seed	/tp	checked
7,262,091	48	-8679320568792204866	-29237064 -34 -16563598	biome (the game: 7,262,043 in ice caves, 48 in deep frozen ocean)
6,965,705	579	4894822981679674548	-29000477 -18 20291750	biome (the game: 6,965,094 in ice caves)
6,517,118	112	-6899656154343219561	-10968151 -14 9204333	biome (the game: 6,517,006 in ice caves)
6,307,406	474	-4925207455282363138	2261232 -19 -19662947	
6,139,235	69	-1871708421544817732	12413921 -20 13341334	

Most ice caves in one cavern

The cavern's blocks in ice caves cells; the rest of it is under other biomes.

ice caves blocks	cavern	share	seed	/tp	checked
12,384,363	21,567,192	57%	1470925652344157552	-394869 -7 -25052170	biome (the game: 12,383,117 in ice caves)
11,259,151	16,944,403	66%	-6280996129646790608	23174992 -27 -13455742	biome (the game: 11,258,490 in ice caves)
10,749,041	15,045,002	71%	8948611172058311402	-28707826 -19 -11259071	
10,287,804	16,468,719	62%	768550052334521	6961044 6 27103692	
10,287,711	12,390,760	83%	742690065862557	-23448166 -18 -18889505	biome (the game: 10,286,986 in ice caves)
10,218,039	11,962,958	85%	8467329850617655169	-2666164 -15 -28103172	
10,155,619	14,846,973	68%	309948833997542484	10707847 2 -5260834	
9,995,914	10,940,218	91%	48091135910896022022	5910355 -10 15774817	

6. What bounds a record



Left: the share of exact tests that stay inside the ice caves, by the size of the scanner's cluster. Middle: where the core of the failures of 700+ points first left the ice caves, and the climate number at fault. Right: the same share for clusters of 700+ points, by the area of the patch of ice caves around them.

A record is a cave sealed inside one patch of ice caves, and the larger the cave the rarer that is. Of the scanner's clusters, 47% of those of 450 to 500 points stay inside the ice caves in the exact test, 15% of those of 600 to 650, 6% at 700 to 800 and 3.7% at 800 to 900 (a point is 8,400 blocks). The ones that leave do so by the weirdness (68%; continentalness 18%, depth 9%, temperature 6%) and at the floor (55% between y -64 and -45, where under an ocean a weirdness of 0.25 to 0.3 is needed and the caves are widest).

The patch is barely larger than the cavern. Around a candidate, the columns that are ice caves from y -56 to 16 form a patch of 0.37 to 0.85 km^2 (10 to 90%; median 0.55), with 80% of its edge set by the weirdness (`icelobe`). A square of 512 blocks inside such a patch is rare: 22 among 3.2 million such columns of 2,400 gated worlds (`icepatch`), while a cavern of 5 to 6 million blocks is 800 to 1,000 blocks long. Among the caverns that pass, size does not go with the patch's area (they hold 7 to 14 blocks of cavern a column of it), but the chance to pass does: 1 to 2% of the clusters of 700+ points in patches under 0.5 km^2, 5 to 7% from 0.65.

Nothing at seed level is left. Among 128 caverns of 4.3M+ no layer of 29 shows an effect (largest 2.7 sigma); the count of a seed's large caverns is Poisson; and the weirdness octave of 256-block cells, which could have been wrong for large patches, is right (mid-cell: +0.23 a stack on patches 256 wide, +0.48 +- 0.23 on those 512 wide). The gate's windows are within 5% of each layer's best.

What moved the tail was a leak, not a gate. Testing the nearly pure clusters took the caverns of 5M+ from 0.23 to 0.90 per 1,000 seeds and the e-fold above 4M from 0.23M to 0.30M +- 0.02. Wider versions add nothing: 900 exact tests of large clusters with 2 to 5% of their samples outside gave nothing over 2.8M. Left to do is compute, and spending it on the largest only: with 700 fine looks, the formula from 560 points and the exact test from 530 pure or 650 nearly pure samples, a seed takes two thirds of the time and 46 of the 49 known caverns of 4.7M+ are still found (the policy of the last runs, box4 and prod6: their counts are complete from 4.7M only).

The strict rule. At the top, three caverns in four whose core is ice caves throughout have a few blocks of their 3-block rim outside. They are listed separately.

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7. Files

`icephase.cpp` (phases, world statistics, the CPU gate), `icegate.cu` (the gate on a card), `icepairs.cpp` (the scanner), `icebiome.cpp`, `iceformula.cpp`, `icemeasure.cpp`, `seed.sh`, `gaterun.sh`, `results.py`; `measure_snapshot.sh` and `verify_ice.py` for the checks against the snapshot; `pairtest.sh`, `recall.sh`, `toprecall.sh`, `indexfit.py`, `gatefit.py` for comparing settings on the same seeds; `tailwhy.py`, `icelobe.cpp`, `icepatch.cpp` for section 6; `fleet/` for the rented box (set-up, start, spending guard); `runs/` for every run; `docs/plots.py` and `docs/data/` for these figures. `README.md` has the measurements behind each number here.